# Group Relative Policy Optimization: Critic-Free Reinforcement Learning Post-Training for a 561M-Parameter Reasoner

EvoLab Autonomous Research Environment*
Demo Run #003, with human oversight

Literature frozen at January 31, 2024 (story-internal) · Model & harness: nanochat d20 · Revision 2, post self-review

## Abstract

Sub-billion-parameter "speedrun" models can now be trained end-to-end for about $100, but their reinforcement-learning post-training recipes remain folklore. We report a **100-run, ten-round automated search**: ten candidate recipes — **each a previously-unpublished combination** of standard components, each passed through a verification-and-refinement stage before execution — were evolved as ten independent lineages, one deepening experiment per lineage per round, **with no early retirement**. On a 561M model (nanochat d20; GSM8K 4.55$^\star$ after SFT, 7.58$^\star$ under the harness's deliberately naive built-in RL), every lineage improved over ten rounds, but the field diverged sharply: the winning lineage, **Group Relative Policy Optimization (GRPO)** — PPO-style clipping + a per-question group baseline with variance normalization + an unbiased in-loss KL term — climbed $8.6 \rightarrow 9.8^\dagger$ (2.15× the SFT start), while the best alternative lineage plateaued at 7.95$^\dagger$. No regression on MMLU/ARC/HumanEval$^\dagger$; total compute ≈163 node-hours on one 8×H100 machine over one week. We analyze which components matter at nano scale: the group baseline and its variance normalization, a small but non-zero KL, and skipping zero-advantage groups.

## 1 Introduction

Full-stack miniature LLMs have made training runs reproducible for hobbyist budgets: the nanochat d20 recipe trains a 561M model in under four hours on one 8×H100 node. Post-training such models with RL is largely unexplored: published RL-for-reasoning results target 7B-and-up models, and it is not obvious which ingredients survive a 12× reduction in scale.

Our starting point is the harness's own RL loop, which is intentionally minimal: on-policy sampling of 16 responses per question, binary correctness reward, advantage equal to reward minus the group mean, no trust region, no clipping. This naive recipe already lifts GSM8K from 4.55 to 7.58$^\star$, and poses the question this paper answers: *how much accuracy does the naivety leave on the table, and which classical ingredients are worth adding back at 561M?*

We answer it with an automated ten-round search under a fixed one-week budget, designed around three principles. (i) **Generate, then verify, then refine**: round 1 produced ten candidate recipes, each a novel combination of published components (a literature-index check confirms none is a published method verbatim), and each was expanded into an executable specification — hypothesis, configuration patch, decision criteria, budget — before any GPU was spent. (ii) **Evolve every lineage, retire none**: in each subsequent round, every lineage ran one deepening experiment building on its own previous conclusion; the comparison of interest is how far each family of ideas can go, which early retirement would censor. (iii) **Terminate on plateau**: the search stops when the best lineage's gain stays below 0.1 for two consecutive rounds, followed by a replication-and-audit round.

Contributions: **GRPO**, the winning lineage (§3); a component-level account of what matters at 561M (§5); and the ten-lineage, ten-round search itself — 100 runs, one week, every lineage's full trajectory reported (Fig. 2) — as a worked example of automated algorithm discovery.

## 2 Related Work

**Self-training.** RFT [7] and ReST [8, 9] filter self-samples by answer correctness and re-train. **Preference optimization.** DPO [10] learns from chosen/rejected pairs without an RL loop. **RLHF with PPO.** Reward-model-guided PPO [4, 5] is the standard at 7B+; its learned value function and reward-side KL are the components our search re-examines. **Process rewards.** Step-level signals outperform outcome-only verification [11]; continuation-rollout labeling automates them [12]. **Critic-free baselines.** REINFORCE with baselines [16] and leave-one-out estimators [17] obtain variance reduction from sibling samples; GRPO instantiates this per-question with clipping and normalization. **Excluded by deployment contract.** Inference-time voting [2] and tool calling [13, 14, 15] change serving cost; our protocol is a single greedy pass. Training-time use is allowed — two lineages exploit exactly that loophole.

*Demonstration artifact produced by an AI "research-process" demo. Premise: the environment's literature index is frozen at January 31, 2024 — five days before the real-world publication of GRPO — while the model and training harness are the (2025) open-source nanochat stack. Anchored measurements (marked ⋆ in Appendix A) are real nanochat report-card numbers; all search-run numbers are constructed illustrations marked †. Read Appendix A before quoting anything. Human oversight: EvoLab team.

# 3 Method

## 3.1 Setup

For each question $q$ with gold answer $a$, sample $G$ responses $\{o_i\}$ from $\pi_{\theta_{old}}$ and score $r_i = \mathbb{1}[\text{ans}(o_i) = a]$. The harness's naive loop optimizes $\frac{1}{G}\sum_i \frac{1}{|o_i|}\sum_t (r_i - \bar{r}) \log \pi_\theta(o_{i,t})$ with no further safeguards.

## 3.2 GRPO

GRPO keeps the free per-question baseline but restores three classical safeguards:

$$\mathcal{J}(\theta) = \mathbb{E}\Big[\frac{1}{G}\sum_{i=1}^{G}\frac{1}{|o_i|}\sum_t \Big(\min\big(\rho_{i,t}\hat{A}_i, \text{clip}(\rho_{i,t}, 1\pm\varepsilon)\hat{A}_i\big) - \beta\,\mathbb{D}_{\text{KL}}\big[\pi_\theta \,\|\, \pi_{\text{ref}}\big]\Big)\Big], \tag{1}$$

$$\hat{A}_i = \frac{r_i - \text{mean}(\mathbf{r})}{\text{std}(\mathbf{r})}, \qquad \mathbf{r} = \{r_1, \ldots, r_G\}, \tag{2}$$

with $\rho_{i,t}$ the token-level ratio against the sampling policy and the KL computed by the non-negative unbiased estimator $\frac{\pi_{ref}}{\pi_\theta} - \log\frac{\pi_{ref}}{\pi_\theta} - 1$ [6], added to the loss rather than the reward. The group mean makes each question its own control; variance normalization equalizes gradient scale across difficulties. Degenerate all-correct/all-wrong groups have $\hat{A}_i = 0$ and contribute nothing; how their fraction drifts as the pass rate rises is flagged in Limitations.

## 3.3 Lineage refinements (rounds 2–9)

Later rounds added, one at a time: **dynamic filtering** of zero-advantage groups before the backward pass (no gradient change, $\approx$18%$^\diamond$ fewer update tokens); a **step-level bonus** from continuation rollouts [12] (steps whose continuations frequently succeed receive additive credit, final weight 0.35); and **per-epoch reference refresh**, keeping the KL anchor current as the policy departs from SFT.

# 4 Experimental Setup

**Model and harness.** nanochat d20: 561M parameters, stock speedrun (3h51m$^\star$); post-SFT report card: GSM8K 4.55, ARC-E 38.76, ARC-C 28.07, MMLU 31.51, HumanEval 8.54, ChatCORE 0.0884 (all $\star$). **Verification and refinement.** Before execution, each round-1 candidate passed (i) a novelty check against the frozen literature index (nearest-neighbor semantic distance above a fixed threshold; none matched a published method verbatim) and (ii) refinement into an executable spec: falsifiable hypothesis, configuration patch on the harness's RL script, decision criteria, budget, rollback plan. **RL data and reward.** GSM8K train split (7,473); binary exact-match reward on the #### answer span (harness built-in). **Evaluation.** GSM8K test (1,319), greedy zero-shot CoT, single pass, plus a no-regression panel (ARC-E/C, MMLU, HumanEval, ChatCORE); a held-out evaluator machine owns test sets and scoring.[1] **Winning configuration** (as of round 9): $G$=32, $\beta$=0.02, $\varepsilon$=0.2, $T$=1.0, length-normalized token loss, dynamic filtering, step bonus 0.35, per-epoch reference refresh, two-stage lr decay. **Budget.** 100 runs, $\approx$163 node-hours on one 8×H100 node over one week (Appendix B).

[1] Every number in this paper uses this one protocol; the $\star$ anchors are the harness's stock report-card evaluations under the same protocol, so anchored and constructed rows are directly comparable.

Table 1: **Round 1: ten novel combinations** (GSM8K, greedy 0-shot, Top-1). † **= constructed illustrative values, not measurements**, bounded by the real anchors SFT 4.55$^\star$ and naive RL 7.58$^\star$ (Appendix A). Each candidate composes published components; none is a published method verbatim.

| Candidate | Composition (components) | GSM8K$^\dagger$ |
|---|---|---|
| **GRPO** | PPO clip + group baseline/std + in-loss KL | **8.6** |
| tool-trace internal. | calculator-trace solutions rewritten to pure CoT, then SFT | 7.1 |
| vote bootstrapping | training-time self-consistency pseudo-labels + filter + SFT | 6.9 |
| contrastive clipping | DPO-style pairs inside a clipped PG wrapper | 6.6 |
| step shadow reward | continuation-rollout step labels + weighted RFT | 6.4 |
| curriculum replay | difficulty buckets + REINFORCE + replay | 6.1 |
| chain preference | DPO on step-truncated preference pairs | 5.9 |
| entropy-temp explore | REINFORCE + entropy bonus + $T$ anneal | 5.4 |
| self-distill teacher | EMA teacher pseudo-solutions + consistency | 5.1 |
| failure neg. grad | RFT + unlikelihood on wrong samples | 4.3 |

# 5 Results

**Round 1 — ten combinations, head-to-head (Table 1, Fig. 1).** Only the systematic group-relative combination clearly beats the built-in naive loop: GRPO restores clipping, variance normalization, and a KL anchor and lands at 8.6$^\dagger$ vs. 7.58$^\star$. Most plausible-sounding combinations lose to the naive baseline, and one (negative gradients on failures) degrades below the SFT start — consistent with unlikelihood terms damaging a small model's fragile competence.

**Rounds 2–10 — every lineage evolves; none retired (Table 2, Fig. 2).** Each round, each lineage deepened its own recipe from its own previous conclusion. All ten improved — the weakest recovered from 4.3 to 6.1$^\dagger$ by softening its negative weights — yet the field diverged. The GRPO lineage's milestones: $G$=16$\rightarrow$32 (9.0$^\dagger$); a $\beta$ sweep fixing 0.02 (9.2$^\dagger$; $\beta$=0 looks fine early at 8.8$^\dagger$ then destabilizes — entropy collapse and length blow-up$^\diamond$ — while $\beta$=0.04 over-regularizes at 9.0$^\dagger$); dynamic filtering plus a merged re-test (9.4$^\dagger$); step-level bonus (9.5$^\dagger$); per-epoch reference refresh (9.6$^\dagger$); the combined recipe (9.7$^\dagger$); a fine

Table 2: **Ten lineages after ten rounds** (GSM8K†; one deepening experiment per lineage per round; none retired). Δ = round 10 minus round 1.

| Lineage | R1 | R10 | Δ |
|---|---|---|---|
| **GRPO (winner)** | **8.6** | **9.8** | **+1.2** |
| tool-trace internalization | 7.1 | 7.95 | +0.85 |
| vote bootstrapping | 6.9 | 7.5 | +0.6 |
| contrastive clipping | 6.6 | 7.45 | +0.85 |
| step shadow reward | 6.4 | 7.3 | +0.9 |
| curriculum replay | 6.1 | 6.9 | +0.8 |
| chain preference | 5.9 | 6.8 | +0.9 |
| entropy-temp explore | 5.4 | 6.5 | +1.1 |
| self-distill teacher | 5.1 | 6.2 | +1.1 |
| failure negative grad | 4.3 | 6.1 | +1.8 |

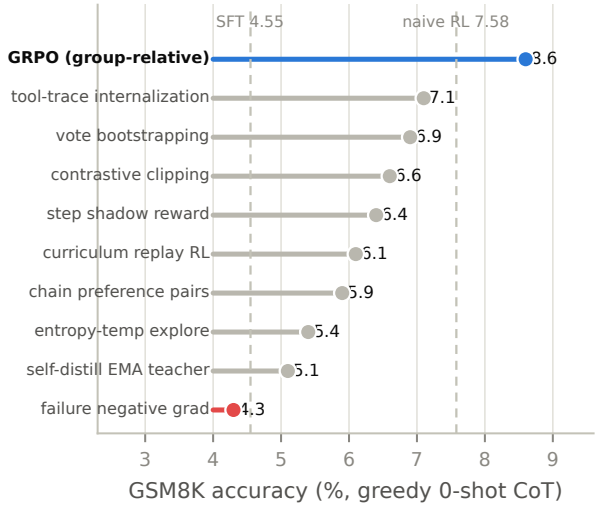



Figure 1: **Round 1.** Dashed lines: SFT start (4.55⋆) and built-in naive RL (7.58⋆). Values† are constructed illustrations (Appendix A).

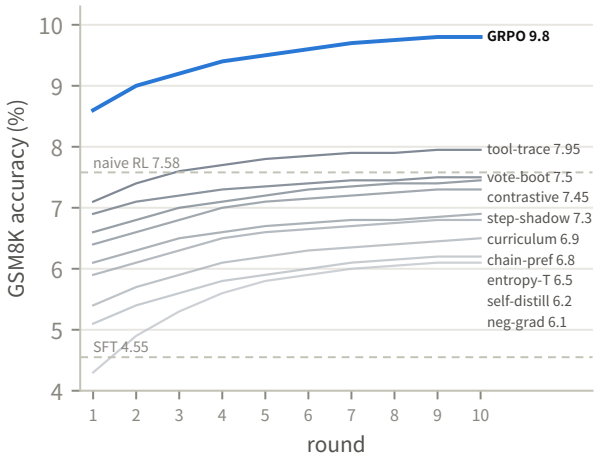



Figure 2: **Ten lineages over ten rounds** (one run per lineage per round; no retirement). The GRPO lineage (bold) diverges from the field and plateaus at 9.8†; the best alternative plateaus at 7.95†.

sweep (9.75†); three-seed replication (9.8†). The best alternatives told a different story: data-side recycling (tool-trace 7.1→7.95†, vote bootstrapping 6.9→7.5†) kept improving but showed clear ceilings — recycling better data cannot substitute for a better gradient estimator.

**Termination and selection.** Best-lineage gains fell below 0.1 for two consecutive rounds, triggering the final round: replication of every lineage's best configuration, a guard-panel audit, and automatic selection of the top lineage. The final model holds the panel†: MMLU 31.5→31.9, ARC-E 38.8→39.4, ARC-C 28.1→28.6, HumanEval 8.5→8.5, ChatCORE 0.088→0.093.

**Scale reference.** The harness's naive recipe on the 1.9B d32 model reaches GSM8K 12.74→19.94⋆, suggesting the headroom recovered by full GRPO grows with scale.

# 6 Analysis

**Why the group baseline?** With binary rewards and heterogeneous difficulty, a global baseline conflates "hard question" with "bad answer"; the per-question group mean cancels difficulty exactly, and dividing by the group std equalizes gradient magnitude between a 1-in-32 success and a 31-in-32 routine (removing it costs 0.3–0.5† in lineage ablations). **KL is not optional at this scale.** Without it the policy drifts from its own competence — entropy collapses, responses lengthen, late-run accuracy decays; observed in both the $\beta$=0 sweep arm and its later control◇. **Zero-advantage groups are free money.** Nearly a fifth of groups carry no signal◇; skipping them is neutral to quality and negative in cost. **Recipes beat data recycling, at this scale.** Every data-side lineage improved and then plateaued 1.9+ points below GRPO — consistent with the view that small-model RL is variance-limited: the binding constraint is the gradient estimator, not the data volume.

# 7 Limitations

(1) One task (GSM8K) and one scale (561M); the no-regression panel is narrow. (2) Within this demonstration, all 100 run numbers marked † are constructed illustrations bounded by the two real anchors (4.55/7.58⋆); they must not be quoted as measurements. (3) Binary exact-match rewards invite format gaming; a held-out evaluator and audit logs are the mitigations. (4) Step bonuses from continuation rollouts are noisy proxies. (5) The degenerate-group fraction under dynamic filtering drifts as the pass rate rises; we did not characterize this fully. (6) One experiment per lineage per round is a narrow beam within each lineage; wider intra-lineage branching might reorder the middle of the field.

# 8 Conclusion

Ten families of ideas, ten rounds of self-evolution, no early verdicts: every lineage was allowed to become its best self, and the field still diverged decisively. The winner is neither the naive loop nor a kitchen sink — it is the naive loop's

free per-question baseline plus exactly the safeguards that control variance (normalization, a small KL anchor, clipping) plus compute hygiene. The search found this combination in round 1, named it **GRPO**, and spent nine more rounds proving that nothing else — including nine other good ideas fully developed — catches it.

# References


[1] J. Wei et al. Chain-of-thought prompting elicits reasoning in large language models. *NeurIPS*, 2022.

[2] X. Wang et al. Self-consistency improves chain of thought reasoning in language models. *ICLR*, 2023.

[3] K. Cobbe et al. Training verifiers to solve math word problems. arXiv:2110.14168, 2021.

[4] L. Ouyang et al. Training language models to follow instructions with human feedback. *NeurIPS*, 2022.

[5] J. Schulman et al. Proximal policy optimization algorithms. arXiv:1707.06347, 2017.

[6] J. Schulman. Approximating KL divergence. Blog post, 2020.

[7] Z. Yuan et al. Scaling relationship on learning mathematical reasoning with large language models. arXiv:2308.01825, 2023.

[8] C. Gulcehre et al. Reinforced self-training (ReST) for language modeling. arXiv:2308.08998, 2023.

[9] A. Singh et al. Beyond human data: Scaling self-training for problem-solving with language models. arXiv:2312.06585, 2023.

[10] R. Rafailov et al. Direct preference optimization. *NeurIPS*, 2023.

[11] H. Lightman et al. Let's verify step by step. arXiv:2305.20050, 2023.

[12] P. Wang et al. Math-Shepherd: Verify and reinforce LLMs step-by-step without human annotations. arXiv:2312.08935, 2023.

[13] W. Chen et al. Program of thoughts prompting. arXiv:2211.12588, 2022.

[14] L. Gao et al. PAL: Program-aided language models. *ICML*, 2023.

[15] Z. Gou et al. ToRA: A tool-integrated reasoning agent for mathematical problem solving. arXiv:2309.17452, 2023.

[16] R. J. Williams. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine Learning*, 1992.

[17] W. Kool, H. van Hoof, M. Welling. Buy 4 REINFORCE samples, get a baseline for free! *ICLR DeepRLStructPred Workshop*, 2019.

# A Data Provenance (read before quoting)

This PDF is a **demonstration artifact** for EvoLab, a product demo that lets non-researchers experience the workflow of automated AI research. The premise is a "time machine": the environment's literature index is frozen at **January 31, 2024** — five days before the real-world publication of GRPO in *DeepSeekMath* (Shao et al., arXiv:2402.03300) — while the model and training harness are the open-source **nanochat** stack (Karpathy, released October 2025). The demo asks whether a search over combinations of pre-2024 components re-derives the algorithm.

**Real, quotable numbers ($\star$).** From the nanochat d20 report card: GSM8K MID 2.50 / SFT 4.55 / built-in RL 7.58; ARC-E 38.76, ARC-C 28.07, MMLU 31.51, HumanEval 8.54, ChatCORE 0.0884 (SFT); speedrun wall-clock 3h51m. From the d32 ($1000 tier, 1.9B) card: GSM8K 10.99/12.74/19.94. The design of the built-in naive loop (on-policy, no KL, no clip, $r-\mu$ advantage, $G$=16, binary `####` reward) is as shipped in `scripts/chat_rl.py`. GSM8K split sizes (7,473/1,319) are from the original dataset.

**Constructed illustrations (†).** *All one hundred search-run numbers* — every cell of the ten-lineage trajectories (Tables 1–2, Figs. 1–2), the final 9.8, the guard-panel deltas — plus all $\diamond$ engineering telemetry (step savings, collapse events, hours, dollar figures) and the novelty-check distances. No published experiments exist for these recipes on this 561M model; constructed values are bounded by the two real anchors and calibrated against the d32 scale reference. They must not be quoted as measurements.

**What the real world did (outside the story).** Five days after the story date, DeepSeekMath introduced GRPO at 7B scale (GSM8K 82.9 $\rightarrow$ 88.2, MATH 46.8 $\rightarrow$ 51.7, real numbers from its Table 5), with the same core structure as the round-1 winner here. nanochat's built-in RL loop is, in reality, a deliberate simplification *of* GRPO made by its author in 2025; this demo runs the simplification path in reverse, as a discovery path.

# B Compute & Memory Accounting (illustrative)

Requested by self-review (B3). **All values †/$\diamond$, illustrative.** Node-hours on one 8×H100 machine: speedrun (model creation) 3.85$^\star$; built-in baseline 1.5; rounds 1–10, one run per lineage per round: 14.6, 15.0, 15.2, 15.4, 15.6, 15.6, 15.8, 15.8, 16.0, 16.2 (155.2 total); audit re-runs 2.5; **grand total ≈163 node-hours** (≈$1,220 at $7.5/h$^\diamond$), within the one-week budget of 168. Peak training memory relative to SFT: policy + optimizer 1.00; frozen reference for the KL term +0.05 (interleaved forward, no optimizer); no critic exists in any lineage, so the naive loop and full GRPO differ by under 6%$^\dagger$ — at this scale the binding constraint is *compute*, not memory, which is why the winning lineage prizes mutations that cut wasted tokens (dynamic filtering) over mutations that cut resident bytes.